

Lecture 1: Course Introduction

Is ChatGPT Conscious? 

PSYC 51.07: Models of Language and Communication

Dr. Jeremy R. Manning

Week 1 - Day 1

Today's Journey

Welcome! 🙌

What is this course about?

We'll explore how machines can understand and generate human language:

- 🤖 Building conversational agents from scratch
- 🧠 Understanding how language relates to thought
- 💻 Hands-on programming with real models
- 🤔 Critical thinking about AI consciousness and capabilities
- 🔬 Research applications in cognitive science

This course is experiential! 🎯

You'll learn by **doing**: coding, experimenting, discussing, and researching.

Course Structure

What we'll build:

1. ELIZA (pattern matching)
2. SPAM classifier
3. Wikipedia analyzer
4. Customer service chatbot
5. GPT model

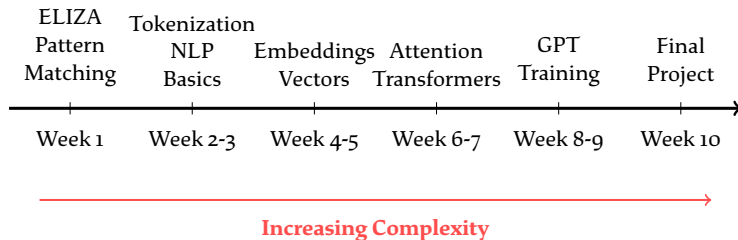
Grading:

-  5 Problem Sets: 75%
-  Final Project: 25%

Tools we'll use:

- Google Colaboratory
- HuggingFace Models
- GitHub
- Discord

Course Roadmap



The Journey

We'll start with simple pattern matching and build up to state-of-the-art transformer models!

The Big Questions

Throughout this course, we'll grapple with:

1. Can machines truly *understand* language?
2. What is the relationship between language and thought?
3. How do statistical patterns capture meaning?
4. Can AI be conscious? Should we care?
5. What can language models tell us about human cognition?



Think about it!

These aren't just philosophical questions—they're at the heart of cognitive science and AI research!

Discussion: Is ChatGPT Conscious?

Discussion

Let's start with a provocative question:

Is ChatGPT conscious?

Think about:

- What does "conscious" even mean?
- How would we test for consciousness?
- Does it matter if ChatGPT *seems* conscious?
- What evidence would convince you either way?

What is Consciousness?

Some perspectives:

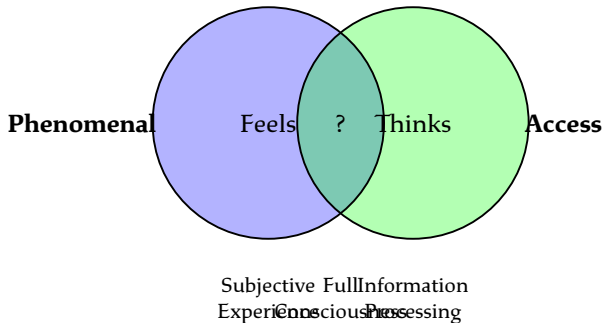
- **Phenomenal consciousness:** Subjective experience ("what it's like")
- **Access consciousness:** Information available for reasoning
- **Self-awareness:** Knowledge of one's own mental states
- **Sentience:** Ability to feel/perceive



Think about it!

If ChatGPT says "I feel happy," does it actually *feel* anything?
How is this different from a parrot saying "I'm hungry"?

Types of Consciousness



The Question

Does ChatGPT have phenomenal consciousness, access consciousness, both, or neither?

The Hard Problem

Why is this so difficult?

We can't directly observe consciousness—even in other humans!

With humans, we assume consciousness because:

- We share similar biology
- We have our own conscious experience
- They behave in ways consistent with having experiences

But with AI:

- ❌ Completely different "biology" (silicon vs. neurons)
- ❌ No shared evolutionary history
- ✅ Can produce human-like behavior
- ? Does the substrate matter?

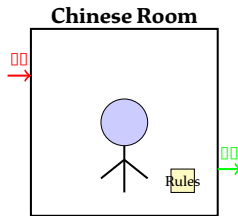
The Chinese Room Argument

John Searle (1980)

Thought experiment about understanding vs. simulation

The setup:

- Person in room with Chinese symbols
- Has rule book for manipulating symbols
- Receives questions in Chinese
- Follows rules, produces answers
- Appears to understand Chinese



Think about it!

Does the person *understand* Chinese?

Important Note

Most cognitive scientists and AI researchers agree: **Current LLMs are not conscious.**

Why not?

- No sensory-motor grounding in the world
- No persistent self-model or goals
- No phenomenal experience (as far as we can tell)
- Pattern matching $\neq$ understanding

Think about it!

But this raises another question: Is consciousness *necessary* for language? Or can you have sophisticated language without consciousness?

Discussion

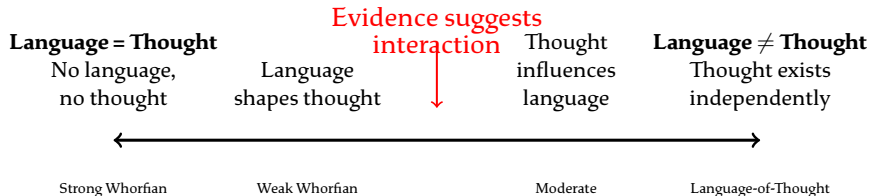
Do you need language to think?
Does language *shape* how you think?

Two extreme positions:

1. **Language = Thought:** All thinking happens in language
 - Implies: No language, no complex thought
2. **Language $\neq$ Thought:** Language is just a tool for communication
 - Implies: Rich mental life exists independently of language

The truth is likely somewhere in between...

The Language-Thought Spectrum



Evidence: The Language Network

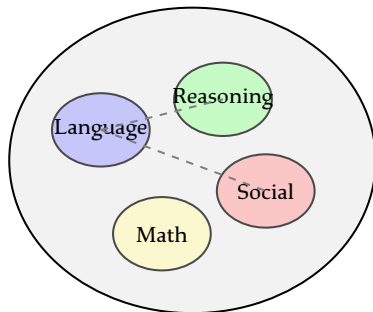
Fedorenko et al. (2024) - Nature

The language network as a natural kind within the broader landscape of the human brain

Key findings:

- The brain has a **specialized language network**
- This network is distinct from:
 - General reasoning/problem-solving
 - Mathematical thinking
 - Social cognition
 - Musical processing
- Language network activates for linguistic tasks, but not for non-linguistic thinking

Implication: Language and thought are *separable* in the brain!



Distinct but connected networks

Key Insight

Language is a specialized system, not the basis of all thought!

Evidence: Language Shapes Perception

Lupyan et al. (2020) - Trends in Cognitive Sciences

Effects of language on visual perception

Key findings:

- Having words for things affects how we *see* them
- Language can:
 - Speed up visual search
 - Alter color perception
 - Influence object categorization
 - Modulate attention
- Example: Russian speakers (with different words for light/dark blue) perceive blue shades more distinctly

Implication: Language doesn't just describe thought—it *shapes* it!

The Russian Blue Example

English Speakers



"blue"

Slower to
distinguish
shades

Russian Speakers



"sinii" (dark blue) "golubiy" (light blue)

Faster to
distinguish
shades

Different categories!



 Think about it!

Language categories affect *perception*, not just description!

So What About LLMs?

Think about it!

If language and thought are separable...

Can an LLM have sophisticated language *without* sophisticated thought?

Can it have sophisticated thought *without* language-like representations?

Current evidence suggests:

- ✓ LLMs are *very good* at language patterns
- ? LLMs may have some internal "representations" of concepts
- ✗ LLMs lack grounding in sensory-motor experience
- ✗ LLMs don't have persistent goals or self-models
- ✗ No evidence of phenomenal consciousness

The Grounding Problem

Symbol Grounding

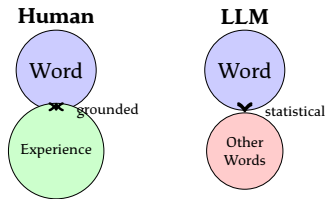
How do symbols (words) get their meaning?

Humans:

- See, touch, taste objects
- Experience causes and effects
- Act in the world
- Meanings grounded in experience

LLMs:

- Only see text
- No sensory experience
- No physical interaction
- Meanings from text patterns



Question

Is statistical grounding "good enough"?

Philosophy of This Course

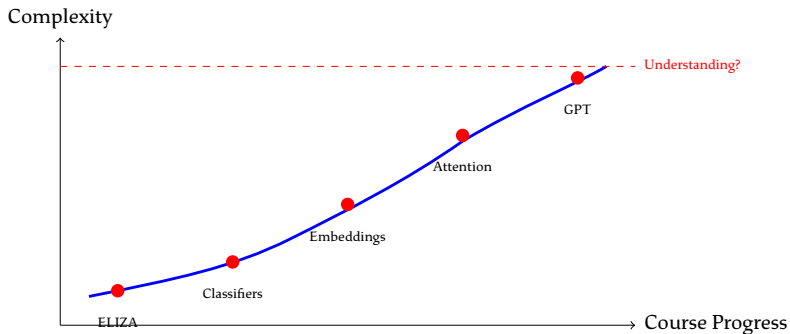
We'll build language models *from scratch* to understand:

1. What they can do
2. What they can't do
3. Why they work
4. What they reveal about language and cognition

Critical perspective:

- Question assumptions about "understanding"
- Appreciate both capabilities and limitations
- Think about implications for cognitive science
- Consider ethical and social dimensions

From Simple to Complex



Discussion

At what point (if any) does pattern matching become understanding?

HuggingFace: Your Learning Companion 🤗

What is HuggingFace?

A platform for sharing and using machine learning models, particularly for NLP.

We'll use HuggingFace for:

- 📖 **Course materials:** Free NLP course
huggingface.co/course
- 🤖 **Pre-trained models:** Download and use state-of-the-art models
- 📊 **Datasets:** Access curated datasets for training
- 🛠️ **Tools:** Tokenizers, trainers, and utilities

Assignment

Create a free HuggingFace account and complete Chapter 1 of the NLP course!

Lecture 2 (Next class): Pattern Matching & ELIZA

- String manipulation fundamentals
- Regular expressions
- Pattern matching techniques
- Introduction to ELIZA

Lecture 3: ELIZA Implementation & The ELIZA Effect

- Complete ELIZA architecture
- Implementing the chatbot
- The ELIZA effect and its implications
- Assignment 1 details

Readings for This Week

Required Readings

1. **Weizenbaum (1966)**: ELIZA — A Computer Program For the Study of Natural Language Communication Between Man and Machine [\[PDF\]](#)
2. **Fedorenko et al. (2024)**: The language network as a natural kind within the broader landscape of the human brain [\[Nature\]](#)
3. **Lupyan et al. (2020)**: Effects of language on visual perception [\[TICS\]](#)

Tip: Start with Weizenbaum—it will help you understand the fundamentals!

Key Takeaways

1. **Consciousness is complex:** Multiple types, hard to define and test
2. **Language $\neq$ Thought:** But they interact in interesting ways
3. **LLMs aren't conscious:** They're sophisticated pattern matchers
4. **Grounding matters:** Meaning comes from experience, not just statistics
5. **Critical thinking is essential:** Don't confuse behavior with understanding
6. **We'll build to understand:** Hands-on experience reveals true capabilities

The Journey Begins

This course will challenge your assumptions about intelligence, language, and mind!

Let's discuss!

Contact Information:

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💬 Discord: <https://discord.gg/sftEk9Ygdw>

🏢 Office Hours: By appointment (Moore Hall 349)

See you next class! 🚀